

Exterior Lighting



NOTE: *The puddle lamps are listed and diagnosed in Section 417-02.*

The exterior lighting system consists of the following components:

- Headlamp assemblies
- Ballasts (HID only)
- Fog lamps
- LED marker lamps (Raptor only)
- Exterior mirrors
- High mounted stoplamp
- Rear lamp assemblies
- License plate lamps
- Headlamp switch
- Stoplamp switch
- Multifunction switch
- Light sensor
- Body Control Module (BCM) (located on the RH side cowl)
- Steering Column Control Module (SCCM) (located on the steering column)

Battery Saver

NOTE: *Time-out is 1 minute if the vehicle has less than 80 km (50 miles).*

NOTE: *The battery saver does **not** control the parking lamps if the headlamp switch is in the PARKING LAMPS ON position.*

To save battery voltage, the BCM provides automatic shut-off of the headlamps and parking lamps after a time-out period when the ignition is off. A timer in the BCM starts when any of the following occur:

- the ignition state changes to off
- any door becomes ajar while the ignition is off
- an unlock button of the Remote Keyless Entry (RKE) transmitter is pressed while the ignition is off
- a valid keypad code is entered while the ignition is off
- the instrument panel dimmer switch is used to turn the courtesy lamps on while the ignition is off

When 10 minutes have elapsed, the BCM automatically shuts off voltage to the lamps. The timer restarts (voltage is restored if the BCM is in battery saver mode) if any of the following occurs:

- the ignition changes out of off
- any door becomes ajar
- the unlock button of the RKE transmitter is pressed
- a valid keypad code is entered
- the instrument panel dimmer switch is pressed

Lamp Assembly Condensation

Exterior lamps are vented to accommodate normal changes in pressure. Condensation can be a natural by product of this design. When moist air enters the lamp assembly through the vents, there is a possibility that condensation can occur if the temperature is cold. When normal condensation occurs, a thin mist forms on the interior of the lens. The thin mist eventually

clears and exits through the vents during normal operation. The amount of time it takes to clear the lens of acceptable mist varies with ambient humidity and lamp types. Normal condensation clears from any lamp in 48 hours under dry conditions.

Do **not** replace a lamp assembly with acceptable levels of condensation such as:

- presence of thin mist (no streaks, drip marks or droplets are present)
- fine mist covers less than 50% of the lens

Examples of unacceptable moisture (usually caused by a lamp housing leak):

- water puddling inside the lamp
- large water droplets, drip marks or streaks present on the interior of the lens

Headlamps

The halogen headlamp system is a dual-beam pattern system and consists of replaceable low/high beam bulbs.

The **HID** headlamp system is a dual-beam pattern system. It consists of a single replaceable **HID** bulb in each headlamp assembly.

The **BCM** supplies voltage to the headlamps when demanded. If the headlamps are on when the ignition switch is turned off, the battery saver feature turns the headlamps off after 10 minutes have elapsed.

Daytime Running Lamps (DRL)

NOTE: *The Daytime Running Lamps (DRL) are a programmable parameter for this vehicle.*

If equipped with halogen headlamps, the **DRL** system utilizes the existing circuitry and components from the headlamp low beam system. The **DRL** system operates the low beam headlamps at a reduced intensity.

If equipped with **HID** headlamps, the **DRL** system utilizes the existing circuitry and components from the front turn lamp system.

The **DRL** are activated when the following conditions are met:

- the ignition is in run
- the headlamps have not been turned on by the autolamp system or the headlamp switch
- the transmission is not in park

Autolamps

The autolamp system provides light sensitive automatic on/off control of the exterior lamps. The autolamp system keeps the exterior lamps on for a preselected period of time after the ignition is turned off (20 seconds is the factory default setting). The preselected time lapse is adjustable up to approximately 3 minutes. To change the autolamps time delay, refer to Autolamps Time Delay Adjustment in this section.

Headlamps On With Wipers On Feature

When the headlamp switch is in the AUTOLAMPS ON position and the front wipers are on for more than 3 seconds (except during a mist wipe or while the wipers are on to clear washer fluid during a wash condition), the exterior lamps turn on. The exterior lamps turn off when the ignition is changed to off or accessory, the headlamp switch is placed in the OFF position, or the front wipers are off for more than 30 seconds. The exception to this is when the exterior lights are on because of darkness determined by the light sensor.

Stoplamps

The stoplamp switch, located on the brake pedal assembly, activates the rear stoplamps and the high mounted stoplamp when the brake pedal is applied. The rear lamps are combination stop/turn lamps.

Turn Signal/Hazard Lamps

When the multifunction switch is placed in the LH or RH TURN position, the corresponding turn lamps cycle on and off approximately 80 times per minute. If a front or rear turn bulb is inoperative, the remaining turn lamps cycle on and off approximately 160 times per minute.

The multifunction switch is a momentary contact switch, with 2 detents for the LH turn position and 2 detents for the RH turn position. When placed in the first detent and released, the corresponding turn signals flash 3 times and turn off. When the multifunction switch is moved to the second detent and released, the turn signal flashes until the steering wheel is turned in the opposite direction, the multifunction switch is moved to the second detent a second time or the vehicle travels farther than 3.2 km (2 miles).

NOTE: *On vehicles equipped with the 6.2L engine the steering wheel rotation sensor is located on the end of the steering column near the intermediate shaft. For all other applications, the steering wheel rotation sensor is part of the EPAS.*

The hazard flasher switch is located in the center instrument panel finish panel (integrated into the Passenger Air Bag Deactivation (PAD)). When the hazard function is active, all the turn lamps cycle on and off approximately 80 times per minute. The hazard flasher lamp switch is a momentary contact switch.

The front turn signal indicator lamps are integrated into the headlamp assemblies.

The rear lamps share functionality with the stoplamps.

Parking Lamps

The parking lamps illuminate when requested from the headlamp switch or the autolamp system.

The front parking and side lamps are integrated into the headlamp assemblies.

When the vehicle is equipped with power telescoping exterior mirrors, the mirrors are equipped with exterior mirror marker lamps.

The rear parking lamps are located in the rear lamp assemblies.

Vehicles equipped with the Raptor option also have LED marker lamps located in the grille and at each corner of the vehicle.

Fog Lamps

The fog lamp switch is integrated into the headlamp switch. The fog lamps can be turned on when the ignition switch is in run or accessory, the parking lamps are on, and the high beams are off.

Reversing Lamps

When the transmission is placed in reverse, the reversing lamps are illuminated. The reversing lamps are located within the rear lamp assemblies.

Trailer Tow

The Battery Junction Box (BJB) supplies voltage to the trailer tow stop/turn, parking and reversing lamps. When the vehicle's rear stop/turn, parking lamps or reversing are illuminated, voltage is supplied to a corresponding trailer tow relay located in the BJB.

The trailer tow battery charge relay provides voltage to the trailer tow connector when the ignition switch is in run.

The vehicle can be equipped with a factory manufactured Trailer Brake Control (TBC) module. For diagnostic information, refer to Section 206-10.

On vehicles not equipped with a factory manufactured TBC module, the vehicle comes equipped with circuitry to support an aftermarket TBC module.